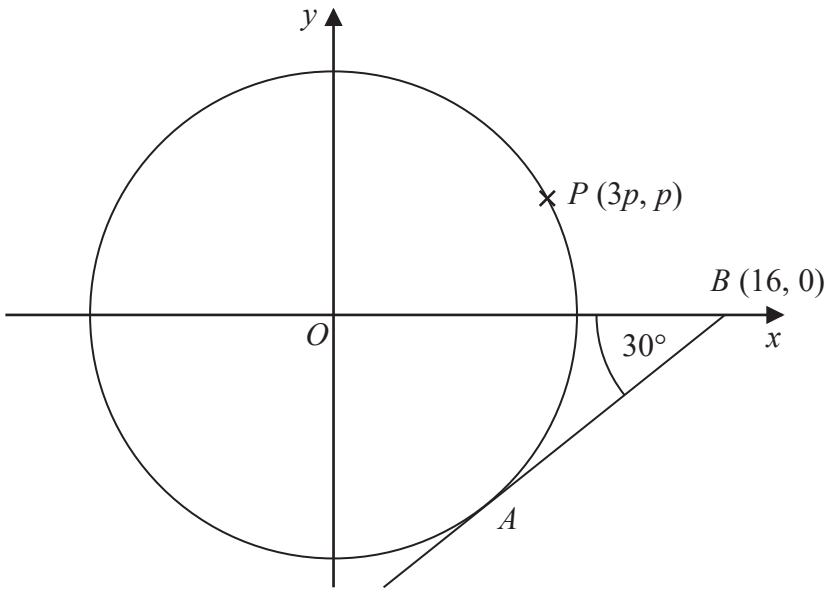
22 The diagram shows a circle, centre O.



AB is the tangent to the circle at the point A.
Angle $OBA = 30°$

Point B has coordinates (16, 0)
Point P has coordinates $(3p, p)$

Find the value of p.
Give your answer correct to 1 decimal place.
You must show all your working.

OA is perpendicular to tangent AB.

$OA = 16 \sin 30° = 8.$

1 mark
Radius OA = 8.

$P(3p,p)$ lies on circle of radius 8:

1 mark
Use $x^2+y^2=r^2$.

$(3p)^2 + p^2 = 8^2 \rightarrow 10p^2 = 64.$

1 mark
Substitute (3p,p).

$p = \sqrt{6.4} = 2.5298... \approx 2.5$ (1 d.p.).

1 mark
p = 2.5 to 1 d.p.

$p =$ 2.5

(Total for Question 22 is 4 marks)